

Checkpoint 12: defect-subspace curvature and the singular survivor

1. Candidate-proof status

There is still no complete proof, and no dimension-free conclusion is claimed at this checkpoint. The principal positive development is a new defect-subspace Poincare theorem which has passed an independent clean-room reproof. Three attempts to feed it into known localization machinery expose one sharply isolated obstruction: a posterior may retain a single weak direction selected from the same future noise that selected its covariance. The displacement and signed-distance routes independently reach the same geometric obstruction as a high-rank family of almost-translations or long normal cells whose global incidence has not yet been controlled.

All constants below are numerical and independent of the dimension.

2. Audited curvature outside a weak subspace

Let $\mathbb{R}^n = F \oplus E$, let $\mu = e^{-V} dx$ be log-concave, and assume distributionally that

$$D^2V \succeq \kappa P_E. \quad (2.1)$$

Writing $\bar{\mu} = (P_F)_\# \mu$, the new theorem is

$$\boxed{C_P(\mu) \leq \kappa^{-1} + 2(C_P(\bar{\mu}) + \kappa^{-1}) \leq 3(C_P(\bar{\mu}) + \kappa^{-1})}. \quad (2.2)$$

For $F = \mathbb{R}u$, the one-dimensional log-concave estimate gives

$$C_P(\mu) \leq 96 (\kappa^{-1} + \text{Var}_\mu \langle X, u \rangle). \quad (2.3)$$

The proof is not a formal appeal to Brascamp–Lieb. Conditional transport velocities $\mathcal{W}_z$ give the reinforced Prekopa inequality

$$D^2U(z) \succeq \kappa G(z), \quad G(z) = \mathbb{E}_z \mathcal{W}_z^* \mathcal{W}_z, \quad (2.4)$$

and matrix Cauchy–Schwarz controls the derivative of the conditional expectation with the weight $(I + G)^{-1}$. The load-bearing marginal estimate is the proved exact-one-form inequality

$$\text{Var}_\eta h \leq 2 \int \langle (D^2U + C_P(\eta)^{-1}I)^{-1} \nabla h, \nabla h \rangle d\eta. \quad (2.5)$$

It follows from the scalar spectral gap on the closure of gradients, the one-form intertwining identity, and the Helffer–Sjostrand variational formula. Gaussian convolution preserves (2.1) with $\kappa_\varepsilon^{-1} = \kappa^{-1} + \varepsilon$, and tensorization gives $C_P(\bar{\mu} * \gamma_\varepsilon) \leq C_P(\bar{\mu}) + \varepsilon$. Thus (2.2) holds for distributional convex potentials and intrinsic affine supports.

The independent audit in `audit_rank_defect_lichnerowicz.md` rederived the reinforced Prekopa inequality, the operator-domain and reducing-subspace steps in (2.5), the matrix comparison, the nonsmooth convolution passage, and the affine-support cases. It found no hidden dimension factor or unproved Poincare input.

3. Ordinary localization does not amplify the theorem

Let

$$\mathcal{K}_d = \sup\{C_P(\nu) : \nu \text{ isotropic log-concave, } \dim \text{aff } \nu \leq d\}.$$

Inserting (2.2) into covariance-whitened stochastic localization and using only the density-martingale and trace identities gives the exact raw recurrence

$$\mathcal{K}_n \leq C \left(\frac{n}{m} + n\mathcal{K}_m \right). \quad (3.1)$$

A spectrum-adaptive water-filling stop gives

$$\mathcal{K}_n \leq C \left[\left(\frac{n}{m} \right)^2 + n \mathcal{K}_m \right]. \quad (3.2)$$

Both are weaker than the elementary trace estimate. Even granting the counterfactual missing assertion that the weak marginal has covariance operator norm $O(1)$, the resulting recurrence

$$\mathcal{K}_n \leq C(n/m + \mathcal{K}_m) \quad (3.3)$$

cannot improve $\mathcal{K}_m \lesssim \log m$: if $m \leq n/\log n$, the curvature term is at least logarithmic, while otherwise $\log m \asymp \log n$. Thus the theorem alone does not even meet M1 through the standard trace bootstrap. A successful localization must generate constant curvature outside $n^{o(1)}$ weak directions while also controlling the covariance of that weak marginal, not merely its dimension.

4. Bounded mass-preserving localization and its precise gate

For a fixed balanced set S , write

$$A_t = \text{Cov}_{\mu_t} X, \quad b_t = \text{Cov}_{\mu_t}(1_S, X), \quad u_t = b_t/|b_t|.$$

On every zero-signal-free interval the bounded controller

$$C_t = P_t = I - u_t u_t^T \quad (4.1)$$

preserves $\mu_t(S)$ pathwise. Its accumulated quadratic tilt is

$$B_t = tI - Q_t, \quad Q_t = \int_0^t u_s u_s^T ds, \quad \lambda_2(B_t) \geq t/2. \quad (4.2)$$

At $t = 1$, (2.3), Buser–Ledoux, and perimeter averaging reduce the desired estimate to

$$\boxed{\mathbb{E} \left(1 + \text{Var}_{\mu_1} \langle X, w_1 \rangle \right)^{-1/2} \geq c,} \quad (4.3)$$

where w_1 is the weakest eigenvector of B_1 . The mixture identity $\mathbb{E} A_1 \preceq I$ applies only to deterministic or past-measurable directions; it gives merely $\mathbb{E} w_1^T A_1 w_1 \leq n$ for this future-selected direction. An explicit adaptive endpoint ensemble achieves equality at scale n , so covariance bookkeeping alone cannot prove (4.3).

The hard convention at $b = 0$ is not well posed: for a one-dimensional Gaussian and a centred interval it contradicts the occupation-density formula. The continuous soft controller

$$C_\varepsilon(b) = I - \frac{bb^T}{|b|^2 + \varepsilon} \quad (4.4)$$

removes this existence defect. It obeys

$$|C_\varepsilon b|^2 = \frac{\varepsilon^2 |b|^2}{(|b|^2 + \varepsilon)^2} \leq \frac{\varepsilon}{4}, \quad (4.5)$$

so the balanced set mass remains in a fixed central interval with constant probability for small fixed ε . Explicitly,

$$\mathbb{P}\{|M_t - \tfrac{1}{2}| > \tfrac{1}{4}\} \leq 4\varepsilon t, \quad \mathbb{P}\{\sup_{s \leq t} |M_s - \tfrac{1}{2}| > \tfrac{1}{4}\} \leq 16\varepsilon t. \quad (4.6)$$

Moreover $C_\varepsilon^2 = I - \alpha uu^T$, $0 \leq \alpha \leq 1$, so all but one eigenvalue of B_1 remain at least $1/2$. This repairs the zero-signal coefficient defect and mass survival on the audited stopped flow, but it does not yet repair (4.3).

The generic-identity obstruction is now realized by an actual projected filter, not only by a static endpoint ensemble. Let μ_n be uniform on $\{\pm\sqrt{n}e_i : 1 \leq i \leq n\}$, and let E be the positive atoms. The exact feedback has $|b_t| \geq 1/2$, hence never meets the zero-signal singularity. With failure probabilities at most

$$992\sqrt{\log n/n} \quad \text{and} \quad 16 \log n/n,$$

the occupation line aligns with the terminal signal and its posterior variance is at least $3n/16$. Consequently the left side of (4.3) is $O(\sqrt{\log n/n})$. This law is isotropic but not log-concave, so it is a rigorous mechanism no-go, not a KLS counterexample.

A static distinction supplied by log-concavity is also proved. For every one-dimensional log-concave Z and every binary label Y ,

$$\mathbb{E} \text{Var}(Z \mid Y) \geq c_0 \text{Var} Z. \quad (4.7)$$

Thus complementary posterior halves cannot collapse to two atoms. Critical linear tilts of the isotropic ℓ_1 ball and of an exponential marginal nevertheless produce genuine long continuous needles. The live log-concavity-only target is the orientation-convexification tail bound

$$\mathbb{P}\{\text{Var}_{\mu_1}\langle X, v_1 \rangle \geq R, \rho_1 \leq r\} \leq C(r + R^{-1/2}), \quad (4.8)$$

where ρ_1 is the accumulated curvature in the terminal occupation line. It is sufficient for (4.3), but is not proved.

The remaining deterministic dichotomy is sharp. If $1 - \lambda_{\max}(Q_1)$ is bounded below, the terminal measure is fully strongly log-concave. Otherwise u_t is $L^2(dt)$ -close to one final line. The unresolved step is to replace that future-selected line by a past-measurable anchor without losing a trace factor. The median maximum of independent shifted exponentials has angular quadratic-variation rate $\Theta((\log n)^2)$ and is the principal stress test for every such anchor argument.

5. Displacement midpoint: bounded strain, unbounded translation rank

For $\mu_+ = 21_E\mu$, $\mu_- = 21_{E^c}\mu$, and the Brenier map $T : \mu_+ \rightarrow \mu_-$, the midpoint law $\nu_{1/2} = ((I + T)/2)_{\#}\mu_+$ satisfies

$$\nu_{1/2} \leq 2\mu, \quad (5.1)$$

and the exact entropy identity

$$\log 2 - \text{Ent}_\mu(\nu_{1/2}) = \mathbb{E} \left[\frac{V(X) + V(TX)}{2} - V\left(\frac{X + TX}{2}\right) + \log \det \frac{I + DT}{2(DT)^{1/2}} \right]. \quad (5.2)$$

The covariance identity is

$$\text{Cov}(\nu_{1/2}) = I - \frac{1}{4}K, \quad K = \mathbb{E}[(T - X) \otimes (T - X)] \preceq 4I. \quad (5.3)$$

The matrix deficit gives the dimension-free Cayley strain budget

$$\mathbb{E} \left\| \frac{DT - I}{DT + I} \right\|_{HS}^2 \leq 2 \log 2. \quad (5.4)$$

It does not see singular Hessian mass or piecewise translations. In the exact zero-strain branch, $D_a T = I$ on the source does not imply that $T - I$ is the gradient of a convex function; a one-dimensional two-cell example has a downward displacement jump across the target gap. Under the additional convex polyhedral displacement hypothesis, translation rank r gives $W_2^2 \leq 4r$, and disjoint midpoint prisms give an exact weighted wall budget. The missing case is a high-rank family of small-strain translations connected through physical gaps or singular walls. Finally,

One high-rank corridor family is now closed. For an isotropic box with a coordinatewise monotone Moreau label, suppose the label is locally constant off coordinate gaps of relative widths δ_j , and the midpoint plateau has half the box volume. Then

$$\prod_j (1 - \delta_j) = \frac{1}{2}, \quad \text{Var}_R b_j \leq \delta_j^2, \quad \sum_j \delta_j \leq \log 2, \quad (5.5)$$

so $\mathbb{E}_R |b - \mathbb{E}b|^2 \leq (\log 2)^2$, apart from the already bounded mean-displacement term. Thus independent orthogonal corridors lose volume multiplicatively while their translation energy adds quadratically. The tensor product that would make the energy add has source mass 2^{-n} ; parity repair reverses some one-dimensional branches and violates Brenier monotonicity. The unproved extension is a mixed-volume analogue of (5.5) for a noncommuting connected power-cell complex.

The mixed-volume extension is now complete whenever the local corridors share one global convex core. If

$$C_0 \subset \dots \subset C_m \subset K, \quad |C_0| = |K|/2, \quad C_{j-1} + [0, v_j] \subset C_j,$$

then isotropy alone gives the projection-thickness bound

$$\frac{|A|}{|P_{u^\perp} A|} \leq \sqrt{12/\alpha} \quad \text{for } A \subset K, \quad |A| = \alpha|K|.$$

The exact extrusion identity $|C + [0, v]| = |C| + |v||P_{v^\perp} C|$ therefore telescopes to

$$\sum_j |v_j| \leq \sqrt{24}. \quad (5.6)$$

The directions need not commute. If the labels lie in the resulting zonotope, then $\text{tr Cov } B \leq 12$ and $\mathbb{E}|B|^2 \leq 13$. Symmetric simplex and crosspolytope fans also close by exact Dirichlet-union and homothetic-core calculations. The remaining premise is local-to-global: arbitrary power walls provide only local facet prisms, not a common half-volume convex core or a laminar sequence of global extrusions.

Finally,

$$\mathcal{W}_2(\mu) \leq 4\sqrt{C_P(\mu)}, \quad D_1(\mu) \leq \frac{1}{2}\mathcal{W}_2(\mu), \quad (5.7)$$

so a universal bound on this displacement is itself KLS-equivalent.

6. Completed singular ray curvature and global incidence

For a balanced normal cell with conditional log-concave density q_y , tail transport W_y , and $\kappa_y = D^2(-\log q_y)$, the correct completed curvature measure on the compactified ray is

$$d\mathcal{C}_y = W_y d\kappa_y + e_y^- \delta_{a_y} + e_y^+ \delta_{b_y}, \quad \mathcal{C}_y([a_y, b_y]) = 2q_y(0). \quad (6.1)$$

After integration, $\mathcal{C} = 2P_\mu(E)$. At a regular medial wall the endpoint contribution is

$$e^{-V}|u^+ - u^-| d\mathcal{H}^{n-1}, \quad (6.2)$$

which is linear BV turning; at a hard support it is $e^{-V}|u \cdot n_K| d\mathcal{H}^{n-1}$. There is no canonical quadratic singular Hessian measure: the transport weight vanishes at the endpoint, and logarithmic smoothing of a Gauss jump drives the weighted quadratic energy to zero.

For a fixed-mass core of cells of scale D , the audited consequences are

$$\mathcal{C}_\mathcal{G} \leq \frac{C}{D}, \quad \text{Cov}_{op}(Y \mid \mathcal{G}) \leq C, \quad \text{tr } Q_\mathcal{G} = 1, \quad \|Q_\mathcal{G}\|_{HS} \leq \frac{C}{D}. \quad (6.3)$$

Hence $\text{rank } Q_\mathcal{G} \gtrsim D^2$. An abstract orthogonal star of long coordinate segments satisfies every numerical constraint with $D \rightarrow \infty$, but is neither log-concave nor the normal congruence of a codimension-one interface.

There is also a new dimension-free local shape estimate. If $s_y = 1/q_y(0)$, the balanced cell contains the normal interval $(-s_y/4, s_y/4)$, and every principal curvature satisfies $|\kappa_j(y)| \leq 4/s_y$. On $|t| \leq s_y/8$, the tail transport is at least $1/4$ and $1/2 \leq 1 + t\kappa_j \leq 3/2$. Therefore its regular normal-Jacobian charge obeys

$$\mathcal{C}_y^{\text{shape}} \geq \frac{s_y}{36} \|S_y\|_{HS}^2, \quad \int_{\mathcal{G}} \|S_y\|_{HS}^2 d\eta(y) \leq \frac{36A}{aD^2}. \quad (6.4)$$

The associated reach inequality

$$|\langle N(y), z - y \rangle| \leq \frac{|z - y|^2}{2L_y}, \quad L_y = s_y/4, \quad (6.5)$$

controls only the normal component of each basepoint chord; direct summation loses the transverse trace. Stable rank also does not produce D^2 direction caps of mass D^{-2} : the realized radial exponential has a uniform spherical Gauss law and exponentially small narrow caps. Prékopa–Leindler fills cross-pair midpoints, but their uncontrolled overlap and transverse trace prevent the naive convexification sum.

A new exact harmless branch has now been proved. If a closed C^2 interface has untruncated regular normal cells for every $t \in \mathbb{R}$, positivity of $\det(I + tS_y)$ forces $S_y = 0$; completeness and absence of cuts make every component a hyperplane. For an isotropic half-mass hyperplane of offset a , Cantelli gives $|a| \leq 1$, and

$$\mathbb{E}|\langle X, u \rangle - a| \leq \sqrt{2}. \quad (6.6)$$

High rank itself cannot be excluded. For the uniform isotropic cube and the parity set, the coordinate-hyperplane fan has balanced cells, $Q = I/n$,

$$J = \frac{\sqrt{3}}{n+1}, \quad P = \frac{n}{2\sqrt{3}}, \quad 2PJ = \frac{n}{n+1}. \quad (6.7)$$

All rigidity is carried by the transition incidence, at exactly the linear-BV scale. The live target is therefore the positive-incidence inequality

$$\mathcal{C}_{\mathcal{G}} + \lambda_{\max}(Q_{\mathcal{G}})^{1/2} \geq c(A, \beta) \quad (6.8)$$

under the realized log-concave normal-congruence hypotheses and the core conditions in (6.3). It holds in the endpoint-free branch and passes the parity fan, radial exponential, and cube-halfspace tests; it is false for the unrealized abstract star. Inequality (6.8) is not yet proved and is currently another exact formulation of the global compatibility gap.

7. Registry after checkpoint 12

Active

1. **Soft bounded localization.** Prove or refute a dimension-free survivor inverse moment after using (4.4), with an adapted-anchor argument that explicitly handles future selection.
2. **Normal-congruence incidence.** Quantify the fact that stable-rank packets of long, almost-orthogonal cells must be convexified by the ambient log-concave density, producing medial/support incidence or basepoint spread.
3. **Zero-gap connectivity.** Convert the exact wall-prism and firm nonexpansiveness identities into an L^2 translation bound, retaining physical target gaps and the nonconvex displacement-potential branch.
4. **Defect theorem applications outside ordinary localization.** Seek a construction whose weak marginal has controlled covariance without invoking a covariance-process estimate equivalent to KLS.

Blocked

1. Ordinary stochastic-localization trace bootstraps, including water-filling, after the explicit recurrences (3.1)–(3.3).
2. Hard projected localization through zero signal, by the occupation-time contradiction.
3. Any claim that $\mathbb{E}A_t \preceq I$ controls an adaptively selected terminal direction.
4. Quadratic singular curvature at medial or hard-support endpoints.
5. Strain-only geometric rigidity, which loses piecewise translations and target gaps.
6. Rank-only ray rigidity, refuted by the radial exponential law and the realized cube parity fan.
7. Same-interface use of Milman’s equivalence to infer $P \lesssim 1/D$; the extremizing set and extremizing Lipschitz function need not coincide.

No item in the active list is being treated as a lemma until it has a dimension-free proof with the approximation, affine-support, and test-model audits requested in the task.